R Cheatsheet

Teaching Week 01 to 12 | The Python sheet carries the same sections in the same order.

This sheet holds the calls this course teaches, in the order it teaches them. The Python cheatsheet carries the same nine sections with the same entries in the same positions, so a student who knows one language can find the other call by reading across. Every example works on `heart.csv`, the table Class 01 opens with, through its `Age`, `MaxHR`, `Cholesterol`, and `HeartDisease` columns. The exceptions are the few lines whose comment names a stand-in (`lookup`, `num_obs`, `num_pred`), which need an object of your own. Load a package once at the top of your script rather than beside every call that needs it.

1 Reading a Table

A lab starts by reading one file into one table.

```
## Read a comma-separated file into a data frame.
heart <- read.csv("heart.csv", stringsAsFactors = TRUE)

## State the separator and the decimal mark when they are not the defaults.
heart <- read.csv("heart.csv", sep = ",", dec = ".")

## Read the first rows only, while you check that the file parses.
read.csv("heart.csv", nrows = 20)

## Treat the strings a source writes for "no value" as missing.
read.csv("heart.csv", na.strings = c("", "NA", "-99"))
```

2 Inspecting It

Confirm the shape, the names, and the type of every column before computing a single statistic.

```
dim(heart)                # rows and columns
names(heart)              # column names
head(heart, 5)            # the first five rows
str(heart)                # type and first values of every column
summary(heart)            # a per-column summary of the whole table
table(heart$HeartDisease) # a count for every level of one column
```

3 Summary Statistics

Centre, spread, quantiles, and correlation, one call each.

```
mean(heart$Age)                       # arithmetic mean
median(heart$Age)                     # middle value
sd(heart$Age)                         # standard deviation, divisor n - 1
var(heart$Age)                        # variance, the square of the above
range(heart$Age)                      # smallest and largest value
quantile(heart$Age, c(0.25, 0.75))    # first and third quartile
IQR(heart$Age)                        # interquartile range, Q3 - Q1
cor(heart[c("Age", "MaxHR")])         # correlation matrix of two columns
```

4 Missing Values

Count what is missing before deciding what to do about it, and never let a default silently decide for you.

```
## heart.csv writes an unmeasured cholesterol as 0, so recode it before counting.
heart$Cholesterol[heart$Cholesterol == 0] <- NA
sum(is.na(heart$Cholesterol))    # how many values of one column are missing
colSums(is.na(heart))            # the same count for every column
which(is.na(heart$Cholesterol))  # the positions of the missing values
complete.cases(heart)            # TRUE where a row has no missing value
clean <- na.omit(heart)          # drop every incomplete row
heart$Cholesterol[is.na(heart$Cholesterol)] <-
  median(heart$Cholesterol, na.rm = TRUE)   # fill with the median instead
mean(heart$Age, na.rm = TRUE)    # a statistic skips NA only when told to
```

5 Plotting

The base call is for a quick look at your screen. The grammar call is for the figure the report cites.

```
library(ggplot2)

hist(heart$Age)                  # base histogram, quick look
boxplot(heart$Age)               # base boxplot, quick look
plot(heart$Age, heart$MaxHR)     # base scatter plot, quick look

ggplot(heart, aes(Age)) +
  geom_histogram(binwidth = 3)   # histogram with a bin width you chose

ggplot(heart, aes(Age, MaxHR, colour = factor(HeartDisease))) +
  geom_point(alpha = 0.6) +      # scatter plot, coloured by the outcome
  labs(x = "Age", y = "Maximum heart rate") +
  theme_minimal()

ggsave("results/age-maxhr.png")  # write the last plot to a file
```

6 Transforming

Filter, select, derive, sort, group, and summarise, in one chain that reads top to bottom.

```r
library(dplyr)

heart |>
  filter(Age >= 40) |>                      # keep the rows that qualify
  select(Age, MaxHR, HeartDisease) |>       # keep the columns you use
  mutate(band = cut(Age, breaks = 4)) |>    # add a derived column
  arrange(desc(MaxHR)) |>                   # sort, largest first
  group_by(HeartDisease) |>                 # form the groups
  summarise(n = n(), mean_age = mean(Age))  # one row per group

scale(heart[c("Age", "MaxHR")])   # centre each column and divide by its sd
factor(heart$HeartDisease)        # read a stored code as a category
## lookup stands in for a second table that shares an id column with heart.
left_join(heart, lookup, by = "id")   # add the columns of a second table
```

7 A Decision Tree

Fit, draw, prune, and predict. The formula names the outcome on the left and the predictors on the right.

```r
library(rpart)
library(rpart.plot)

fit <- rpart(
  factor(HeartDisease) ~ Age + MaxHR,   # outcome ~ predictors
  data   = heart,
  method = "class",                     # "anova" predicts a number instead
  control = rpart.control(cp = 0.01, minsplit = 20, maxdepth = 2)
)

rpart.plot(fit)                         # draw the fitted tree
printcp(fit)                            # complexity table, one row per split
## cp is alpha / R(root) on misclassification risk, not Python's ccp_alpha,
## so the two values differ. Pick a cp between two rows of the table.
fit <- prune(fit, cp = 0.1)             # cut the tree back to a complexity
predict(fit, heart, type = "class")   # the predicted class of every row
```

8 K-Means Clustering

Scale first, because the algorithm measures distance and an unscaled column decides the answer on its own. Fix the seed so the run repeats.

```r
library(cluster)

Z <- scale(heart[c("Age", "MaxHR")])   # comparable units before any distance
set.seed(3141)
km <- kmeans(Z, centers = 3, nstart = 25, iter.max = 100)

km$cluster                        # the cluster label of every row
km$centers                        # the k centres, in scaled units
km$tot.withinss                   # total within-cluster sum of squares
s <- silhouette(km$cluster, dist(Z))   # the silhouette width of every row
mean(s[, 3])                      # the average silhouette width
```

9 Evaluating a Model

Split the data, score the rows the model has not seen, and report the metric the outcome type calls for.

```r
library(caret)

set.seed(3141)
idx   <- sample(nrow(heart), floor(0.7 * nrow(heart)))
train <- heart[idx, ]                # 70 percent to fit on
test  <- heart[-idx, ]               # the rest to be scored on

fit  <- rpart(factor(HeartDisease) ~ Age + MaxHR,
              data = train, method = "class")
pred <- predict(fit, test, type = "class")
obs  <- factor(test$HeartDisease)

table(obs, pred)             # confusion matrix, observed against predicted
mean(pred == obs)            # accuracy, the share the model got right
confusionMatrix(pred, obs, positive = "1")  # all three, with 1 as positive

train(factor(HeartDisease) ~ Age + MaxHR, data = heart, method = "rpart",
      trControl = trainControl(method = "cv", number = 10))   # 10-fold CV

## num_obs, num_pred stand in for observed and predicted values of a number.
postResample(pred = num_pred, obs = num_obs)   # RMSE, R squared, and MAE
R2(num_pred, num_obs, form = "traditional")    # R squared as 1 - SSres/SStot
```